MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEECT205 - CONTINUOUS AND DISCRETE SYSTEMS
(2024 ADMISSION ONWARDS)
QP Code: 25901234

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Part A
Short Answer
Answer any 10 questions
Each question carries 2 marks

1. Write a short note on the IC 741 op-amp.

Pin diagram/ block diagram or any reasonable explanation – 2 marks

2. Which BJT configuration is most popular for amplification and why?

Common emitter configuration – high voltage & current gain – 2 marks

3. What is an Op-amp? Draw the schematic symbol of an op-amp.

Meaningful definition of Op-amp – 1 mark; Symbol – 1 mark

4. State the Barkhausen criteria.

Loop gain = 1 and phase shift = 0° or 360° – 2 marks

5. Which component mainly decides the time constant in an op-amp integrator?

RC components – 2 marks

6. Which component forms the frequency determining network in an op-amp Colpitts oscillator?

LC tank circuit – 2 marks

7. What is meant by a subtractor circuit using an op-amp?

Difference of two input voltages / diagram of / any meaningful explanation– 2 marks

8. What is a counter?

Sequential circuit that counts pulses/ a diagram/ any meaningful explanation – 2 marks

9. Find 1's complement of 10011011.

01100100 – 2 marks

10. Subtract: 1011 − 0101.

0110 – 2 marks (Partially correct , simple error in taking borrow- 1 mark)

11. Simplify: Y = A + AB.

Y = A – 2 marks

12. Importance of logic gates.

Basic building blocks of digital systems or any gate symbols and truth tables – 2 marks

13. Difference between Binary and Hexadecimal.

Binary: base 2 – 1 mark; Hexadecimal: base 16 – 1 mark

OR any examples – 2 marks

14. What are minterms?

Product terms containing all variables OR Examples – 2 marks

Part B
Answer any 6 questions
Each question carries 5 marks

15. Explain RC coupled amplifier with circuit diagram.

Diagram – 3 marks; Explanation– 2 marks

16. If Rf = 100 k ohm and R1 = 10k ohm, calculate the gain for both inverting and

non- inverting configurations of op-amp.

Inverting: - Gain = 10 – 2 marks (Equation- 1 mark, Substitution and answer-1 marks)

Non-inverting: Gain = 11 – 3 marks (Equation- 2 mark, Substitution and answer-1 marks)

17. Define (a) Input Offset Voltage (b) Input Offset Current (c) CMRR (d) Slew Rate

Each definition – marks distributed (total 5)

18. List important features of the 555 Timer and draw the pin diagram.

Features – 3 marks; Diagram – 2 marks

19. Design a High pass filter for the cut off frequency of 2 KHz and draw its circuit diagram and frequency response.

Formula – 1 mark; Calculation – 2 marks; Diagram – 1 mark; Response – 1 mark

20. Compare SR and JK flip-flops.

SR – 1; JK – 1; Comparison – 3 marks

21. Describe the role of truth tables in understanding logic gate operations.

Explanation – 5 marks (examples of logic gates and truth tables)

22. What is the equivalent expression for the Boolean expression x'y'z +yz+ xz?

x′y′z+yz+xz

yz+xz+x′y′z

z(y+x)+ x'y'z

OR any equivalent reduction procedure

Steps – 3 marks; Answer – 2 marks